$S°(298.15\ K)$ = [265.269 ± 5.9] J·K^{-1}·mol^{-1}

$\Delta_f H°(0\ K)$ = −230.56 ± 41.8 kJ·mol^{-1}
$\Delta_f H°(298.15\ K)$ = −232.63 ± 41.8 kJ·mol^{-1}

Vibrational Frequencies and Degeneracies ν, cm^{-1}
[970](1)
[300](1)
[930](1)

Ground State Quantum Weight: [2] σ = [2]
Point Group: [C_{2v}]
Bond Distance: V–O = [1.589] Å
Bond Angle: O–V–O = [110]°
Product of the Moments of Inertia: $I_A I_B I_C$ = [2.858786 × 10^{-115}] g^3·cm^6

Enthalpy of Formation

Four independent mass spectrometric studies lead to values for $\Delta_f H°(VO_2, g, 298.15\ K)$.[1-4] Shchukarev *et al.*[2] studied the composition of the vapor above V_2O_4 during evaporation from the surface of a platinum strip. A 2nd law sublimation heat, $\Delta_{sub}H°$ = 104 ± 2 kcal·mol^{-1}, was reported for the range 1500–1800 K. Assuming the reported value corresponds to 1700 K and using auxiliary data,[5] we calculate $\Delta_f H°(VO_2, g, 298.15\ K)$ = −62.3 ± 2.0 kcal·mol^{-1}. The sublimation was described by 0.5 V_2O_4(cr) = VO_2(g) More recent related work[3] yields a $\Delta_f H°(298.15\ K)$ value approximately 6 kcal·mol^{-1} more positive. This latter work and other investigations are summarized in the following table.

Reference	Reaction	Data Points	T/K	$\Delta_r H°(298.15\ K)$, kcal·mol^{-1} 2nd law	3rd law	Drift cal·K^{-1} mol^{-1}	$\Delta_f H°(298.15\ K)$, kcal·mol^{-1} 2nd law	3rd law
[3]	A	Equation	1673–1818	114.29	114.32	0.02	−56.26	−56.23
[3]	B	Equation	1818–2273	104.90	103.77	−0.58	−54.30	−55.43
[1]	C	2	1948	–	−16.23	–	–	−45.97
[4]	C	1	2270	–	−12.86	–	–	−49.34
[4]	D	1	2270	–	−40.14	–	–	−32.94 (−39.79)**

**Value is based on recent determination by Farber *et al.*[8] for $\Delta_f H°$(AlO, g, 298.15 K).
Reactions: (A) 0.5 V_2O_4(cr) = VO_2(g) (C) V(g) + VO_2(g) = 2 VO(g)
(B) 0.5 V_2O_4(l) = VO_2(g) (D) 2 AlO(g) + V(g) = 2 Al(g) + VO_2(g)

We adopt $\Delta_f H°(298.15\ K)$ = −55.6 kcal·mol^{-1} for VO_2(g). This value is an average of the 2nd and 3rd law results of Frantseva and Semenov.[3] These results are independent of the VO(g) data and show small third law drifts. The results of our analysis for reaction (C) for the data of Berkowitz *et al.*[1] and Farber *et al.*[4] show good agreement in $\Delta_r H°(298.15\ K)$ but lead to values for $\Delta_f H°(298.15\ K)$ which are 6–10 kcal·mol^{-1} less negative than our adopted value. Killingbeck[6] studied the evaporation of V_2O_3 from a molybdenum effusion chamber and reported $\Delta_r H°(298.15\ K)$ = 279.5 ± 5 kcal·mol^{-1} for V_2O_3(cr) = VO(g) + VO_2(g). Using auxiliary data[5] we calculate $\Delta_f H°(298.15\ K)$ = −42.3 ± 5.0 kcal·mol^{-1} for VO_2(g).

The adopted $\Delta_f H°(298.15\ K)$ value corresponds to a dissociation energy $D°$ = 295.5 kcal·mol^{-1} (12.8 eV) for the process VO_2(g) = V(g) + 2 O(g). This $D°$ value is 1.96 times the dissociation energy of VO(g).[5]

Heat Capacity and Entropy

The vibrational frequencies and geometry of VO_2(g) are estimated based on existing and estimated data for related metal dioxides such as TiO_2(g), TaO_2(g) and WO_2(g).[5] The V–O bond distance is assumed to be the same as in VO(g).[5]

The ground state quantum weight of 2 is adopted so as to be consistent with TaO_2(g).[5] This introduces an uncertainty of ±1.38 cal·K^{-1}·mol^{-1} in $S°(298.15\ K)$. The estimation of ν_2 at 300 cm^{-1} results in an uncertainty of 0.72 cal K^{-1} mol^{-1} in $S°(298.15\ K)$ for ±100 cm^{-1}. The estimates for VO_2(g) are such that the 3rd law drifts as indicated in the above table are small. Nevertheless, until more data are available for VO_2(g), the uncertainty in the entropy must be considered as ±1.4 cal·K^{-1}·mol^{-1} for a bent molecule. The free energy functions suggested by Brewer and Rosenblatt[7] are roughly 3–4 cal·K^{-1}·mol^{-1} lower than those adopted here. The main reason for the difference is that Brewer and Rosenblatt[7] treated VO_2(g) as a linear molecule.

The electronic levels of the V^{+4} ion as reported by Moore[9] consist of a ground state with quantum weight four and an excited state at 620 cm^{-1} with a quantum weight six. The next excited state is listed at ~148000 cm^{-1}. Inclusion of an estimated state for VO_2(g) at 620 cm^{-1} with a quantum weight of six would raise $S°(298.15\ K)$ by 1.06 cal·K^{-1}·mol^{-1}. This is within the estimated uncertainty of $S°(298.15\ K)$.

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Enthalpy Reference Temperature = T_r = 298.15 K Standard State Pressure = $p°$ = 0.1 MPa

T/K	$C_p°$	$S°$	$-[G°-H°(T_r)]/T$	$H°-H°(T_r)$	$\Delta_f H°$	$\Delta_f G°$	log K_f
	J·K^{-1}mol^{-1}			kJ·mol^{-1}			
0	0.	0.	INFINITE	−11.254	−230.561	−230.561	INFINITE
100	35.385	222.647	301.443	−7.880	−230.580	−234.796	122.645
200	39.819	248.608	269.153	−4.109	−231.557	−238.670	62.334
250	41.903	257.716	265.981	−2.066	−232.116	−240.383	50.225
298.15	43.907	265.269	265.269	0.	−232.630	−241.928	42.385
300	43.982	265.541	265.270	0.081	−232.650	−241.986	42.134
350	45.916	272.468	265.812	2.330	−233.144	−243.503	36.341
400	47.622	278 714	267.041	4.669	−233.597	−244.951	31.987
450	49.082	284.409	268.659	7.088	−234.017	−246.345	28.595
500	50.311	289.646	270.499	9.573	−234.412	−247.693	25.876
600	52.201	298.996	274.489	14.705	−235.162	−250.278	21 789
700	53.531	307.149	278.585	19.995	−235.902	−252.739	18.860
800	54 485	314.363	282.615	25.398	−236.669	−255.092	16.656
900	55.184	320.822	286.507	30.884	−237.491	−257.347	14.936
1000	55 708	326.665	290.236	36.429	−238.380	−259.506	13.555
1100	56.110	331.994	293.793	42.021	−239.344	−261.572	12.421
1200	56.424	336.891	297.183	47.649	−240.399	−263.547	11.472
1300	56.674	341.417	300.414	53.304	−241.552	−265.430	10.665
1400	56.875	345.625	303.495	58.982	−242.816	−267.220	9.970
1500	57.039	349.554	306.436	64.678	−244.192	−268.916	9.364
1600	57.175	353.240	309.247	70.388	−245.681	−270.516	8.831
1700	57.288	356.710	311.938	76.112	−247.293	−272.020	8.358
1800	57.384	359.987	314.517	81.846	−249.035	−273.425	7.935
1900	57.466	363.092	316.993	87.588	−250.919	−274.729	7.553
2000	57.536	366.041	319.372	93.338	−252.953	−275.931	7.207
2100	57.597	368.850	321.662	99.095	−255.148	−277.026	6.891
2200	57.649	371.531	323.868	104.857	−280.396	−277.908	6.598
2300	57.695	374 094	325.997	110.625	−283.080	−277.735	6.308
2400	57.736	376.551	328.052	116.396	−285.781	−277.445	6.038
2500	57.772	378.908	330.040	122.172	−288.501	−277.042	5.788
2600	57.804	381.175	331.963	127.951	−291.239	−276.530	5.556
2700	57.832	383.357	333.826	133.732	−293.995	−275.912	5.338
2800	57.858	385.461	335.633	139.517	−296.768	−275.191	5.134
2900	57.881	387.491	337.387	145.304	−299.559	−274.372	4.942
3000	57.902	389.454	339.090	151.093	−302.368	−273.455	4.761
3100	57.921	391.353	340.745	156.884	−305.193	−272.444	4.591
3200	57.938	393.192	342.356	162.677	−308.034	−271.343	4.429
3300	57.953	394.975	343.923	168.472	−310.891	−270.153	4.276
3400	57.967	396 705	345.450	174.268	−313.763	−268.875	4.131
3500	57.981	398.386	346.939	180.065	−316.650	−267.513	3.992
3600	57.992	400.020	348.391	185.864	−319.551	−266.068	3.861
3700	58.004	401.609	349.808	191.664	−769.272	−263.342	3.718
3800	58.014	403.156	351.191	197.465	−770.514	−249.651	3.432
3900	58.023	404.663	352.543	203.266	−771.819	−235.927	3.160
4000	58.032	406.132	353.865	209.069	−773.188	−222.169	2.901
4100	58.040	407.565	355.157	214.873	−774.623	−208.376	2.655
4200	58.047	408.964	356.421	220.677	−776.124	−194.547	2.420
4300	58.055	410.330	357.659	226.482	−777.691	−180.681	2.195
4400	58.061	411.664	358.872	232.288	−779.325	−166.779	1.980
4500	58.067	412.969	360.059	238.094	−781.026	−152.839	1.774
4600	58.073	414.245	361.223	243.901	−782.795	−138.860	1.577
4700	58.078	415.494	362.365	249.709	−784.634	−124.841	1.387
4800	58.083	416.717	363.485	255.517	−786.541	−110.783	1.206
4900	58.088	417.915	364.583	261.326	−788.518	−96.684	1.031
5000	58.092	419.089	365.662	267.135	−790.566	−82.544	0.862
5100	58.097	420.239	366.720	272.944	−792.686	−68.362	0.700
5200	58.101	421.367	367.761	278.754	−794.878	−54.140	0.544
5300	58.104	422.474	368.782	284.564	−797.144	−39.872	0.393
5400	58.108	423.560	369.787	290.375	−799.485	−25.563	0.247
5500	58.111	424.626	370.774	296.186	−801.902	−11.210	0.106
5600	58.114	425.673	371.745	301.997	−804.396	3.190	−0.030
5700	58.117	426.702	372.700	307.809	−806.969	17.633	−0.162
5800	58.120	427.713	373.640	313.621	−809.622	32.124	−0.289
5900	58.123	428.706	374.565	319.433	−812.356	46.658	−0.413
6000	58.126	429.683	375.476	325.245	−815.175	61.243	−0.533

PREVIOUS: December 1973 (1 atm) CURRENT. December 1973 (1 bar)